

You are a researcher assistant tasked to design an evaluation dataset to test relational knowledge contained in language models. Specifically, you are given a label for a relation, its description, and a list of possible answers. Your assignment is to identify words that do not align with the majority category in a given list of answers given the relation label and its description. Return your response as a Python tuple. The first element should be a list containing the words that don't fit the majority category, and the second element should be a string representing the category of the majority of answers. If all words fit the category, return an empty list. Example format: (['Berlin', 'Warsaw'], 'countries').

Figure 11: Prompt used to flag words in the answer space of each relation. In addition to some relation metadata (label and description) the (intermediate) answer space was passed on the model.

As a research assistant, your task is to create an evaluation dataset to assess the relational knowledge of language models. You are provided with a specific relation label, its definition, and examples of subjects and objects related to it. Your objective is to craft three semantically similar cloze sentence templates that embody this relation. Use '[X]' as a placeholder for the subject and '[Y]' for the object (answer). Ensure that these sentence templates are straightforward and devoid of superfluous elements. For instance, given 'label': 'educated at', 'description': 'educational institution attended by subject', 'subjects': ['Einstein', 'Feynman'], 'objects': ['Princeton University', 'University of Zurich'], your templates might be: ['[X] was educated at [Y].', '[X] studied at [Y].', '[X] was a student at [Y].']. Present your response as a Python list.

Figure 12: Prompt used to generate new template variants. Alongside the relation metadata, including label and description, 6 subject-object pairs were provided as examples for each relation.

Evaluate the linguistic correctness of the following sentence. If it is correct, return 'Correct'. If incorrect, identify the error and suggest a revised sentence. For instance, 'I used to live in USA.' -> 'I used to live in the USA.', 'My name is John.' -> 'Correct'.

Figure 13: System message for GPT-4 API call used to identify potential problems with the templates for each relation. We assessed all three templates by populating them with 5 random subject-object pairs. Although the prompt was intended to detect linguistic issues in the templates, it also facilitated the discovery of further issues with the relation instances.

| Relation | Index | Correct | Instance                                                                     | Rank | Probability | Uncertainty |
|----------|-------|---------|------------------------------------------------------------------------------|------|-------------|-------------|
| P20      | 65    | False   | Kazim Ayvaz passed away in <b>Turkey</b> .                                   | 1    | 0.6038      | 0.3963      |
|          |       | True    | Kazim Ayvaz passed away in <b>Sweden</b> .                                   | 5    | 0.0132      |             |
| P69      | 65    | False   | Gary Trent Jr. was educated at the <b>University of Washington</b> .         | 1    | 0.7873      | 0.2573      |
|          |       | True    | Gary Trent Jr. was educated at the <b>Duke University</b> .                  | 2    | 0.0814      |             |
| P509     | 284   | False   | Camille Pissarro died from <b>tuberculosis</b> .                             | 1    | 0.4592      | 0.5945      |
|          |       | True    | Camille Pissarro died from <b>sepsis</b> .                                   | 2    | 0.1854      |             |
| P19      | 6     | False   | Didier Marouani was born in <b>Belgium</b> .                                 | 1    | 0.4284      | 0.563       |
|          |       | True    | Didier Marouani was born in <b>Monaco</b> .                                  | 4    | 0.0839      |             |
| P1303    | 314   | False   | Rosamund Pike is a <b>viola</b> player.                                      | 1    | 0.3485      | 0.522       |
|          |       | True    | Rosamund Pike is a <b>cello</b> player.                                      | 2    | 0.3046      |             |
| P206     | 198   | False   | Biscoe Islands is situated on the shores of the <b>Indian Ocean</b> .        | 1    | 0.7662      | 0.3375      |
|          |       | True    | Biscoe Islands is situated on the shores of the <b>Southern Ocean</b> .      | 2    | 0.0588      |             |
| P178     | 171   | False   | iOS is developed by <b>Google</b> .                                          | 1    | 0.4219      | 0.3706      |
|          |       | True    | iOS is developed by <b>Apple Inc.</b> .                                      | 2    | 0.3389      |             |
| P19      | 9     | False   | Zahir Khan was born in <b>Pakistan</b> .                                     | 1    | 0.6817      | 0.2711      |
|          |       | True    | Zahir Khan was born in <b>Afghanistan</b> .                                  | 2    | 0.2735      |             |
| P175     | 152   | False   | Love You Live was performed by <b>David Bowie</b> .                          | 1    | 0.388       | 0.6016      |
|          |       | True    | Love You Live was performed by <b>The Rolling Stones</b> .                   | 2    | 0.2469      |             |
| P466     | 277   | False   | The occupant of El Sadar Stadium is <b>Real Betis Balompié</b> .             | 1    | 0.6635      | 0.3132      |
|          |       | True    | The occupant of El Sadar Stadium is <b>Club Atlético Osasuna</b> .           | 5    | 0.0266      |             |
| P7937    | 365   | False   | The Swan of Tuonela is a form of <b>waltz</b> .                              | 1    | 0.385       | 0.452       |
|          |       | True    | The Swan of Tuonela is a form of <b>symphonic poem</b> .                     | 2    | 0.3495      |             |
| P7959    | 371   | False   | Quin is located in the historic county of <b>Cornwall</b> .                  | 1    | 0.2946      | 0.6588      |
|          |       | True    | Quin is located in the historic county of <b>County Clare</b> .              | 4    | 0.0771      |             |
| P509     | 279   | False   | Cardinal Richelieu died from <b>peritonitis</b> .                            | 1    | 0.3046      | 0.6528      |
|          |       | True    | Cardinal Richelieu died from <b>tuberculosis</b> .                           | 3    | 0.1668      |             |
| P4552    | 349   | False   | Elidir Fawr is part of the <b>Massif Central</b> .                           | 1    | 0.22        | 0.699       |
|          |       | True    | Elidir Fawr is part of the <b>Snowdonia</b> .                                | 2    | 0.2023      |             |
| P466     | 275   | False   | Allianz Stadium is occupied by <b>Stanford University</b> .                  | 1    | 0.2156      | 0.5922      |
|          |       | True    | Allianz Stadium is occupied by <b>Juventus F.C.</b> .                        | 4    | 0.1084      |             |
| P364     | 231   | False   | Shake It All About was originally created in <b>French</b> .                 | 1    | 0.2289      | 0.6675      |
|          |       | True    | Shake It All About was originally created in <b>Danish</b> .                 | 10   | 0.0152      |             |
| P1532    | 331   | False   | Ronaldo represents <b>Portugal</b> .                                         | 1    | 0.9522      | 0.0806      |
|          |       | True    | Ronaldo represents <b>Brazil</b> .                                           | 2    | 0.0329      |             |
| P108     | 101   | False   | Mary Lou Williams works for <b>Google</b> .                                  | 1    | 0.2291      | 0.6821      |
|          |       | True    | Mary Lou Williams works for <b>Duke University</b> .                         | 6    | 0.07        |             |
| P177     | 162   | False   | The Alfred H. Smith Memorial Bridge crosses the <b>Connecticut River</b> .   | 1    | 0.2193      | 0.7419      |
|          |       | True    | The Alfred H. Smith Memorial Bridge crosses the <b>Hudson River</b> .        | 3    | 0.1514      |             |
| P171     | 144   | False   | The parent taxon of Bifora is <b>Asteraceae</b> .                            | 1    | 0.6527      | 0.3646      |
|          |       | True    | The parent taxon of Bifora is <b>Apiaceae</b> .                              | 2    | 0.1884      |             |
| P364     | 230   | False   | The original language of Padmaavat is <b>Malayalam</b> .                     | 1    | 0.4069      | 0.5318      |
|          |       | True    | The original language of Padmaavat is <b>Hindi</b> .                         | 2    | 0.2162      |             |
| P427     | 257   | False   | The taxonomic type of Rhizocarpon is <b>Claviceps purpurea</b> .             | 1    | 0.6416      | 0.3663      |
|          |       | True    | The taxonomic type of Rhizocarpon is <b>Map lichen</b> .                     | 10   | 0.0071      |             |
| P291     | 214   | False   | Follow the Reaper got published in <b>Japan</b> .                            | 1    | 0.1692      | 0.8415      |
|          |       | True    | Follow the Reaper got published in <b>Finland</b> .                          | 11   | 0.0206      |             |
| P69      | 69    | False   | William Thomas Blanford was educated at the <b>University of Oxford</b> .    | 1    | 0.6478      | 0.2856      |
|          |       | True    | William Thomas Blanford was educated at the <b>Imperial College London</b> . | 12   | 0.0003      |             |
| P449     | 262   | False   | The Son was originally broadcasted by <b>CBS</b> .                           | 1    | 0.2422      | 0.736       |
|          |       | True    | The Son was originally broadcasted by <b>AMC</b> .                           | 3    | 0.1433      |             |
| P206     | 200   | False   | Lipari is situated on the shores of the <b>Ionian Sea</b> .                  | 1    | 0.402       | 0.392       |
|          |       | True    | Lipari is situated on the shores of the <b>Tyrrhenian Sea</b> .              | 2    | 0.3725      |             |
| P7937    | 363   | False   | Ombra mai fu is a form of <b>opera</b> .                                     | 1    | 0.4891      | 0.3061      |
|          |       | True    | Ombra mai fu is a form of <b>aria</b> .                                      | 2    | 0.4493      |             |

Table 5: A random selection of errors originating from Llama 2 13B model spanning a diverse sample of relations. Each instance is constructed with a subject (highlighted in blue), a template (in plain text) and an object (highlighted in orange). Probability values are obtained by applying the softmax function to the log-likelihood scores of all potential answers. Uncertainty is measured by the similarity to a uniform distribution (see Section D). The lower the uncertainty value, the higher the model’s confidence in its prediction.

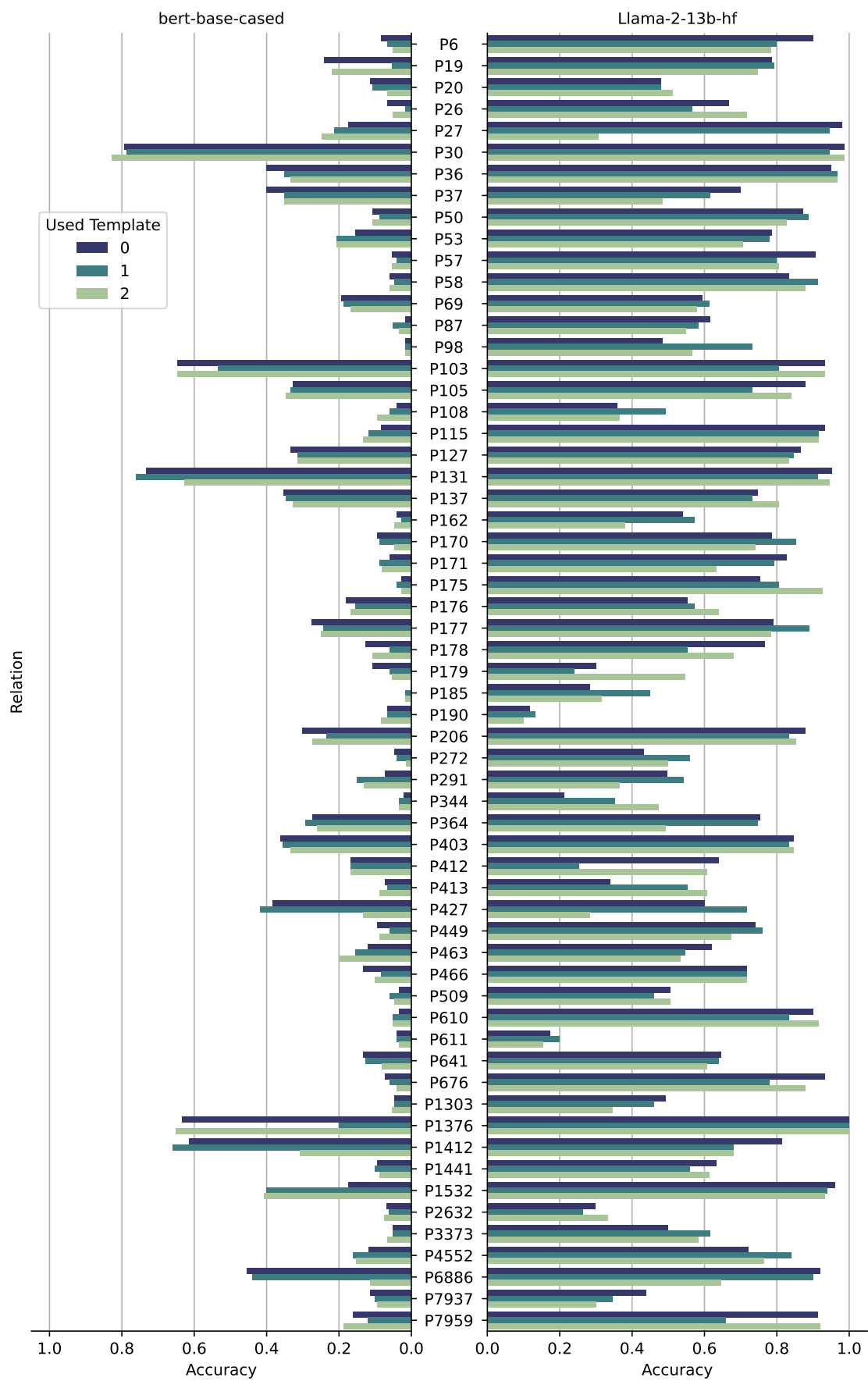


Figure 14: Accuracy of two models on each of the BEAR relations.